Channel-Based Q&A App

Complete Build Checklist · Deadline: April 7, 2026 @ 3:00 PM

Phase 0 — Environment & Repo Setup

0.1 Initialise Project

- [x] ~~Create Git repo and add .gitignore (node_modules, .env, uploads/)~~
- [x] ~~Run: npx create-next-app@latest with App Router selected~~
- [x] ~~Choose TypeScript (recommended) and Tailwind CSS~~
- [x] ~~Verify dev server starts: npm run dev → localhost:3000~~

0.2 Docker Skeleton

- [x] ~~Create Dockerfile at repo root (node:20-alpine base)~~
- [x] ~~Create docker-compose.yml with app + db services~~
- [x] ~~Add named volume for DB data persistence~~
- [x] ~~Expose correct ports (e.g. 3000:3000 app, 5432:5432 Postgres)~~
- [x] ~~Confirm: docker compose up --build starts without errors~~

0.3 Database Choice & Schema

- [x] ~~Choose SQL (PostgreSQL recommended) or NoSQL — document the reason~~
- [x] ~~Create migrations/schema file implementing all 6 entities:~~
  - [x] ~~User (id, passwordHash, displayName, role)~~
  - [x] ~~Channel (id, name UNIQUE, description, createdBy, createdAt)~~
  - [x] ~~Post (id, channelId, authorId, title, body, createdAt)~~
  - [x] ~~Reply (id, postId, parentReplyId?, authorId, body, createdAt)~~
  - [x] ~~Vote (id, userId, targetType, targetId, value, UNIQUE(userId, targetId))~~
  - [x] ~~Attachment (id, targetType, targetId, mimeType, sizeBytes, path/url)~~
- [x] ~~Add indexes on foreign keys and searchable fields~~
- [x] ~~Create seed script with: 1 admin, 2+ users, 3+ channels, posts, replies, votes~~
- [x] ~~Verify seed runs cleanly inside Docker container~~

0.4 Environment Config

- [x] ~~Create .env.example listing ALL required variables (DATABASE_URL, NEXTAUTH_SECRET, etc.)~~
- [x] ~~Add .env to .gitignore — never commit real secrets~~
- [x] ~~Confirm app reads env vars at startup without crashing~~

PART 1 30%

Phase 1 — Basic System (Channels, Posts, Replies, Uploads)

1.1 Landing Page

- [x] ~~Create / route with a clear explanation of what the tool is~~
- [x] ~~Add navigation links to channel list and login/signup~~
- [x] ~~Show empty state when no channels exist~~

1.2 Channels

- [x] ~~GET /api/channels — returns list of all channels~~
- [x] ~~POST /api/channels — creates a channel (auth required later, stub for now)~~
- [x] ~~Channel list page: displays name, description, post count~~
- [x] ~~Channel create form with name (required, unique) and description (optional)~~
- [x] ~~Selecting a channel navigates to its post list~~
- [x] ~~Empty state shown when channel has no posts~~

1.3 Posts

- [x] ~~GET /api/channels/[id]/posts — returns posts for a channel~~
- [x] ~~POST /api/channels/[id]/posts — creates a post~~
- [x] ~~Post list shows: title, author display name, date, reply count~~
- [x] ~~Post create form with title (required) and body (required)~~
- [x] ~~Clicking a post opens its detail view with full body~~

1.4 Replies

- [x] ~~GET /api/posts/[id]/replies — returns flat or nested replies~~
- [x] ~~POST /api/posts/[id]/replies — creates a reply~~
- [x] ~~Reply form visible on post detail page~~
- [x] ~~Replies displayed below the post body~~

1.5 Screenshot Uploads

- [x] ~~File input on post/reply create forms (optional attachment)~~
- [x] ~~Server-side validation: PNG, JPEG/JPG, WebP only~~
- [x] ~~Server-side validation: max 5 MB per file~~
- [x] ~~Validate MIME type AND file extension (not just extension)~~
- [x] ~~Store file safely: filesystem path or DB blob (no raw execution path)~~
- [x] ~~Serve via controlled route (e.g. /api/attachments/[id]) — not direct FS URL~~

- [x] ~~Display uploaded image(s) inline with post/reply~~
  ⚠ *Never serve uploads from the same path they are written to.*

1.6 Data Persistence Check

- [x] ~~Restart Docker containers — data survives restart (volume mounted)~~
- [x] ~~Confirm seed data loads automatically on first start~~

PART 2 **20%**

Phase 2 — Accounts & Admin

2.1 Authentication

- [x] ~~POST /api/auth/signup — hashes password with bcrypt (min 10 rounds), creates user~~
- [x] ~~POST /api/auth/signin — verifies hash, issues session token/cookie~~
- [x] ~~POST /api/auth/signout — destroys session~~
- [x] ~~NEVER store plaintext passwords — verify in DB after sign-up~~
- [x] ~~Implement session (NextAuth.js / iron-session / JWT) — document choice~~
- [x] ~~Add CSRF protection if using cookie sessions~~
  ⚠ *NextAuth.js with credentials provider is the simplest approach for Next.js.*

2.2 Enforce Auth on All Write Actions

- [x] ~~Middleware or per-route guard: create channel → must be logged in~~
- [x] ~~Create post → must be logged in~~
- [x] ~~Create reply → must be logged in~~
- [x] ~~Cast vote → must be logged in~~
- [x] ~~Upload attachment → must be logged in~~
- [x] ~~Return 401 from API for unauthenticated write attempts~~
- [x] ~~Show clear 'Please sign in' message in UI when action is blocked~~

2.3 Display Names

- [x] ~~Author display name shown on every post, reply, and vote~~
- [x] ~~Profile link or tooltip showing author stats (optional but nice)~~

2.4 Admin Powers

- [x] ~~Seed includes at least one user with role = 'admin'~~
- [x] ~~Admin UI: Remove any user (with confirmation dialog)~~
- [x] ~~Admin UI: Remove any channel (with confirmation dialog)~~
- [x] ~~Admin UI: Remove any post (with confirmation dialog)~~

- [x] ~~Admin UI: Remove any reply (with confirmation dialog)~~
- [x] ~~API routes for each admin delete enforce admin role check~~
- [x] ~~Non-admins cannot access admin routes (return 403)~~

2.5 Input Safety

- [x] ~~Validate all inputs server-side: required fields, max lengths~~
- [x] ~~Sanitise / escape output to prevent XSS (React does this by default for JSX — avoid dangerouslySetInnerHTML)~~
- [x] ~~Add basic rate limiting on login endpoint (e.g. 10 req/min per IP)~~
- [x] ~~Add basic rate limiting on post/reply creation~~

PART 3 10%

Phase 3 — Nested Replies & Voting

3.1 Threaded Replies

- [x] ~~Reply form has option to reply to another reply (not just the root post)~~
- [x] ~~Reply stores parentReplyId (null for top-level, reply ID for nested)~~
- [x] ~~API returns replies with nesting info (parentReplyId)~~
- [x] ~~UI renders replies as an indented tree (recursive component)~~
- [x] ~~No hard nesting depth limit — component renders recursively~~
- [x] ~~Within a thread, replies ordered oldest-first~~

3.2 Voting

- [x] ~~Vote API: POST /api/votes — upserts vote (up/down/remove)~~
- [x] ~~Database enforces UNIQUE(userId, targetType, targetId)~~
- [x] ~~Users can change vote: up → down or down → up~~
- [x] ~~Users can remove vote: clicking same state toggles to neutral~~
- [x] ~~Vote counts / net score displayed on posts and replies~~
- [x] ~~Voting requires auth — unauthenticated users see score but cannot vote~~
- [x] ~~Document your decision: can authors vote on their own content?~~

PART 4 10%

Phase 4 — Search

4.1 Search Page

- [x] ~~Dedicated /search page with search bar and filter controls~~
- [x] ~~Results include context snippet + link to source post/channel~~
- [x] ~~Pagination or 'Load More' button on results~~
- [x] ~~Empty state when no results found~~

4.2 Query Types (implement ≥ 5)

- [x] ~~Substring search across post titles and bodies~~
- [x] ~~Substring search across reply bodies~~
- [x] ~~Filter: content by a specific author (username/display name)~~
- [x] ~~Stat query: user with most posts (show count)~~
- [x] ~~Stat query: user with least posts (show count)~~
- [x] ~~Ranking query: highest-voted content~~

- [x] ~~(Optional) Filter by channel~~

4.3 Search Quality

- [x] ~~Database indexes added on searchable columns~~
- [x] ~~Document: which fields are searched + ranking formula used~~

Phase 5 — UX, Accessibility & Polish

5.1 Loading & Error States

- [x] ~~Loading spinner/skeleton shown while fetching channels, posts, replies~~
- [x] ~~Empty state components: no channels, no posts, no replies, no search results~~
- [x] ~~API errors surfaced to user with a readable message (not raw JSON)~~
- [x] ~~Form validation errors displayed inline (not just alert())~~

5.2 Responsive Layout

- [x] ~~Sidebar/channel list collapses or stacks on mobile~~
- [x] ~~Post and reply forms usable on small screens~~
- [x] ~~Images scale correctly on mobile~~

5.3 Accessibility

- [x] ~~All form inputs have associated <label> elements~~
- [x] ~~Primary user flows completable using keyboard only (Tab, Enter, Space)~~
- [x] ~~Buttons and links have visible :focus styles~~
- [x] ~~Status (up-voted, down-voted) conveyed by more than colour alone~~
- [x] ~~Text/background colour contrast ≥ 4.5:1 for body text~~

5.4 Admin UX

- [x] ~~Confirmation dialog before every destructive admin delete~~

Phase 6 — Testing

6.1 Test Report (10+ cases required)

- [x] ~~TC-01: Sign up with valid data → user created, redirected~~
- [x] ~~TC-02: Sign up with duplicate username → clear error shown~~
- [x] ~~TC-03: Sign in with wrong password → access denied~~
- [x] ~~TC-04: Create channel while logged out → 401 returned, UI blocked~~
- [x] ~~TC-05: Create channel while logged in → appears in list~~
- [x] ~~TC-06: Create post in channel → appears with author name~~
- [x] ~~TC-07: Reply to a post → reply shown threaded~~
- [x] ~~TC-08: Reply to a reply → nested correctly~~
- [x] ~~TC-09: Upload non-image file (e.g. .exe) → rejected with error~~
- [x] ~~TC-10: Upload image > 5 MB → rejected with error~~
- [x] ~~TC-11: Upload valid PNG → displays inline with post~~
- [x] ~~TC-12: Upvote a post → score increases by 1~~
- [x] ~~TC-13: Change vote from up to down → score decreases by 2~~
- [x] ~~TC-14: Remove vote → score returns to original~~
- [x] ~~TC-15: Search for keyword present in a post → result appears with link~~
- [x] ~~TC-16: Admin deletes a post → post removed for all users~~
- [x] ~~TC-17: Non-admin attempts admin delete → 403 returned~~
- [x] ~~TC-18: Sign out → write actions blocked until sign in again~~

Phase 7 — Documentation

7.1 README.md

- [x] ~~Step-by-step run instructions (clone → docker compose up → seed → open browser)~~
- [x] ~~List all exposed ports (app, DB)~~
- [x] ~~Admin credentials for seed user~~
- [x] ~~How to run migrations / seed data~~
- [x] ~~Link to demo video~~

7.2 .env.example

- [x] ~~All environment variables listed with placeholder values~~
- [x] ~~No real secrets committed — .env in .gitignore~~

7.3 Design Report (1 page)

- [x] ~~Architecture overview: Next.js routes → API layer → DB~~

- [x] ~~Database choice justification (why SQL/NoSQL)~~
- [x] ~~All API endpoints listed with method, path, auth requirement~~
- [x] ~~Screenshot storage approach explained (filesystem / DB blob / object store)~~
- [x] ~~Key packages listed with one-line justification each~~

7.4 Test Report (1 page)

- [x] ~~All test cases documented (ID, description, steps, expected, actual, pass/fail)~~
- [x] ~~Screenshots or logs as evidence where applicable~~

7.5 Demo Video (10 min)

- [x] ~~Walk through all 4 parts of the rubric~~
- [x] ~~Show Docker startup from a clean clone~~
- [x] ~~Demonstrate admin moderation actions~~
- [x] ~~Demonstrate search with multiple query types~~
- [x] ~~Upload link included in README~~

Phase 8 — Final Smoke Checks Before Submission

8.1 Clean Clone Test

- [x] ~~Clone repo to a fresh directory (no node_modules, no .env)~~
- [x] ~~Copy .env.example → .env and fill in values~~
- [x] ~~Run: docker compose up --build~~
- [x] ~~Confirm migrations/seed run automatically (or follow README steps)~~
- [x] ~~App accessible at documented port with seed data visible~~

8.2 Feature Verification

- [x] ~~Sign up, sign in, sign out all work~~
- [x] ~~Logged-out user cannot create content (401/redirect)~~
- [x] ~~Create channel, post, reply as regular user~~
- [x] ~~Upload image to a post — displays correctly~~
- [x] ~~Nested reply renders in tree view~~
- [x] ~~Vote up/down/remove updates score in real time (or on refresh)~~
- [x] ~~Search returns relevant results with context links~~
- [x] ~~Admin can delete user, channel, post, reply~~

8.3 Security Spot-Checks

- [x] ~~POST to /api/channels with no cookie → 401~~

- [x] ~~POST to /api/admin/users/[id] as non-admin → 403~~
- [x] ~~Upload .php or .exe file → rejected~~
- [x] ~~Upload 6 MB image → rejected~~
- [x] ~~Check DB: passwords stored as hashes (bcrypt $2b$ prefix)~~

8.4 Submission Package

- [x] ~~docker-compose.yml present at repo root~~
- [x] ~~All source files, package.json, and lockfile committed~~
- [x] ~~DB migrations/schema and seed file(s) present~~
- [x] ~~README.md complete with run steps, ports, admin creds~~
- [x] ~~.env.example present with no real secrets~~
- [x] ~~Design Report document included~~
- [x] ~~Test Report document included~~
- [x] ~~Demo video link in README is accessible (not private)~~
- [x] ~~Final commit pushed before April 7, 2026 @ 3:00 PM~~

Total: 100% | Deadline: April 7, 2026 @ 15:00